SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Machine Learning

ASSESSMENT 1 – Analytical Problem solving (5 QUESTIONS)

Course Code:	ITA05	**Course Title:**	**Computer Vision**
CO Assessed:	CO2	**Assessment:**	Assessment 1: Analytical Problem solving
Weightage:	35% of CIA	**Total Marks:**	(5 Questions × 10 Marks)
CO2 Statement:	***Apply image enhancement, filtering, and segmentation techniques to analyze images, detect edges and boundaries, and solve real-world Computer Vision problems.***		
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Section / Batch:	Slot A/2024	**Date:**	21/10/2026

Code of Conduct

I s.srijitha 192471024 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: *srijitha*

QUESTIONS: 5 **Total Marks: 50**

CO2 Assessment Tool-01

Analytical Problem Solving

Q1.A grayscale image contains the following pixel intensity values: 10, 10, 20, 20, 20, 30, 30, and 40. Construct the histogram by calculating the frequency of each intensity value. Calculate the cumulative distribution function (CDF) for the given image. Apply histogram equalization using the formula s = round ((L − 1) × CDF), where L = 256. Determine the new intensity values and compare the contrast distribution before and after equalization.

Answer

Histogram frequencies: Intensity 10: 2 pixels; intensity 20: 3 pixels; intensity 30: 2 pixels; intensity 40: 1 pixel. Total pixels = 8.

CDF calculation: For intensity 10: 2/8 = 0.25. For intensity 20: (2+3)/8 = 5/8 = 0.625. For intensity 30: (2+3+2)/8 = 7/8 = 0.875. For intensity 40: 8/8 = 1.00.

Histogram equalization: Using s = round (255 × CDF): 10 → round (255 × 0.25) = 64; 20 → round (255 × 0.625) = 159; 30 → round(255 × 0.875) = 223; 40 → round(255 × 1.00) = 255.

New intensity sequence: 64, 64, 159, 159, 159, 223, 223, 255.

Comparison: Before equalization, all pixel values lie in the narrow range 10–40, so the image has low contrast. After equalization, the values spread across 64–255, making intensity differences more visible and improving the overall contrast.

Q2.A 3 × 3 image has the intensity values [[10, 20, 30], [20, 100, 20], [10, 20, 30]]. Apply a 3 × 3 averaging filter to calculate the new intensity value of the centre pixel. Replace the centre pixel with the calculated average while keeping the other pixels unchanged. Apply a median filter to the same neighbourhood and calculate the centre pixel value. Compare the results of the averaging and median filters and determine which filter better preserves image details in the presence of noise.

Answer

Averaging filter: Sum of all nine values = 10 + 20 + 30 + 20 + 100 + 20 + 10 + 20 + 30 = 260. Average = 260/9 = 28.89 (approximately 29).

Image after averaging: [[10, 20, 30], [20, 28.89, 20], [10, 20, 30]]. If integer pixel values are required, the centre value is rounded to 29.

Median filter: Sorted values = 10, 10, 20, 20, 20, 20, 30, 30, 100. The middle (fifth) value is 20, so the new centre pixel is 20.

Image after median filtering: [[10, 20, 30], [20, 20, 20], [10, 20, 30]].

Comparison: The averaging filter changes the centre to about 28.89 because the extreme value 100 raises the mean. The median filter changes it to 20 and is less affected by this outlier. Therefore, the median filter generally preserves edges and image details better when removing impulse noise, while averaging tends to blur edges.

Q3. Consider a grayscale image with the intensity matrix [[50, 50, 50], [50, 100, 50], [50, 50, 50]]. Apply the Laplacian filter mask [[0, −1, 0], [−1, 4, −1], [0, −1, 0]] to calculate the response at the centre pixel. Use the calculated response to sharpen the centre pixel by adding it to the original intensity value. Determine the resulting centre pixel intensity after sharpening. Explain how the Laplacian filter enhances intensity transitions and fine details in the image.

Answer

Laplacian response: The mask multiplies the centre by 4 and subtracts its four direct neighbours. Response = (4 × 100) − 50 − 50 − 50 − 50 = 400 − 200 = 200.

Sharpened centre intensity: Sharpened value = original centre intensity + Laplacian response = 100 + 200 = 300. Thus, the mathematical result is 300. Since an 8-bit grayscale pixel can only range from 0 to 255, the stored pixel must be clipped to 255 if the image is kept in 8-bit format.

Explanation: The Laplacian is a second-order derivative filter that responds strongly where intensity changes rapidly. Adding its positive response to the bright centre increases the centre-to-neighbour contrast, emphasizing the local intensity transition and making fine details or sharp features more prominent.

Q4.A grayscale image contains the pixel intensity values [[40, 60, 120], [80, 150, 200], [30, 100, 180]]. Apply global thresholding with a threshold value of 100, assigning 1 to pixels greater than or equal to the threshold and 0 to the remaining pixels. Construct the resulting binary image. Calculate the number of foreground and background pixels after thresholding. Explain how the selected threshold separates objects from the background and identify one limitation of this method.

Answer

Thresholding rule: Assign 1 when intensity ≥ 100; otherwise assign 0.

Binary image: [[0, 0, 1], [0, 1, 1], [0, 1, 1]].

Pixel counts: Foreground pixels (value 1): 5. Background pixels (value 0): 4. Total = 9 pixels.

Explanation and limitation: The threshold separates pixels with intensity 100 or above as foreground from darker pixels as background. This is simple and effective when object and background intensities are distinct. One limitation is that a single global threshold may fail under uneven illumination or when foreground and background intensity ranges overlap.

Q5. A grayscale image has the intensity matrix [[10, 10, 10], [10, 50, 90], [10, 50, 90]]. Apply the Sobel operator using the masks Gx = [[−1, 0, 1], [−2, 0, 2], [−1, 0, 1]] and Gy = [[−1, −2, −1], [0, 0, 0], [1, 2, 1]] to calculate the gradient components at the centre pixel. Calculate the gradient magnitude using $G = \sqrt{(Gx^2 + Gy^2)}$. Determine the dominant edge direction based on the calculated gradient components. Explain how the detected edge information can be used for boundary detection and image segmentation.

Answer

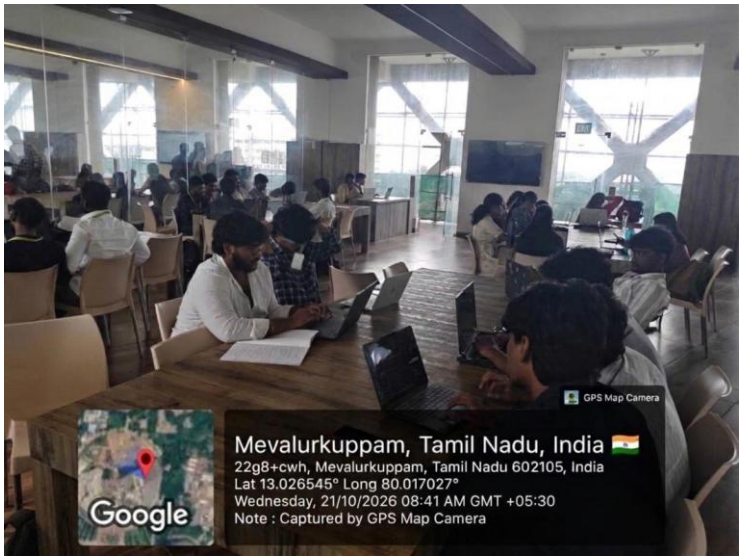
Horizontal gradient component (Gx): Top row: −10 + 10 = 0. Middle row: (−2 × 10) + (2 × 90) = −20 + 180 = 160. Bottom row: −10 + 90 = 80. Therefore, Gx = 0 + 160 + 80 = 240.

Vertical gradient component (Gy): Top row: (−10) + (−20) + (−10) = −40. Middle row contributes 0. Bottom row: 10 + 100 + 90 = 200. Therefore, Gy = −40 + 0 + 200 = 160.

Gradient magnitude: $G = \sqrt{(240^2 + 160^2)} = \sqrt{(57{,}600 + 25{,}600)} = \sqrt{83{,}200} \approx 288.44$.

Dominant direction: Since $|Gx| = 240$ is greater than $|Gy| = 160$, the gradient is stronger in the x direction. The gradient angle is $\theta = \tan^{-1}(Gy/Gx) = \tan^{-1}(160/240) \approx 33.69°$ from the positive x-axis. The edge line itself is perpendicular to the gradient direction.

Use in boundary detection and segmentation: Large gradient magnitudes indicate strong intensity changes and likely object boundaries. Edge pixels can be linked to form contours, which help separate objects from the background or divide an image into regions. Noise can also create false edges, so smoothing or thresholding the gradient may be needed.





Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Content Accuracy	30	Highly accurate, indepth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____